

The table-free latitudinal march

Lean 4 / Mathlib proof of record: `formal/lean/` (v4.34.0-rc2).

Abstract. We record the identities used by the table-free latitudinal march. For spin $s \geq 0$, order $m \geq 0$ and $x = \cos \theta$, the Wigner function is the Jacobi polynomial

$$d_{m,-s}^\ell(\theta) = (-1)^{m+s} 2^{\varepsilon_m} \text{LG2N}(\ell, m) \left(\sin \frac{\theta}{2}\right)^{m+s} \left(\cos \frac{\theta}{2}\right)^{|m-s|} P_{\ell-M}^{(m+s, |m-s|)}(x),$$

with $M = \max(m, s)$ and $\varepsilon_m = \text{sign}(m - s)$ (equal to $+1$ at $m = s$). The kernel marches this polynomial by the three-term recurrence, carries an exact binary exponent, emits a binade-split partial, and may skip polar tiles by ducc0's m_{lim} cutoff. Every identity below is proved in Lean; only the WKB estimate of §8 is not.

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1 Introduction

The transform implemented by `gmaster/_spin_march_pallas.py` is, for each order m and each of four real channels $r_c(\theta)$,

$$A_c(\ell, m) = \sum_{\theta} d_{m,-s}^\ell(\theta) r_c(\theta), \quad \ell = \max(m, s), \dots, L - 1, \quad (1.1)$$

where $L = \ell_{\text{max}} + 1$. The kernel never stores associated Legendre tables. It evaluates $d_{m,-s}^\ell$ from a Jacobi recurrence in compensated float32, with an integer exponent lane, a float32 emit, and an optional polar skip.

This note is the human-readable statement. The proof of record is the Lean 4 project `GMasterMarch`; Appendix A names the theorems. The project has no `sorry`: DLMF 18.9.2

for the explicit Jacobi sum is proved in `JacobiSum.jacobi_three_term` (a telescoping certificate over the summands, §3), the identification of the Jacobi closed form with Wigner’s explicit sum in `WignerJacobi.wignerD_eq_jacobiForm` (§2), and the spin-2 mirror identity in `Mirror.mirror_identity` (§6). The WKB estimate in §8 is asymptotic and has no Lean counterpart.

The argument is organised as follows. §2 substitutes $m' \mapsto m$, $m \mapsto -s$ into Edmonds’s Jacobi form of the small- d matrix. §3 rearranges DLMF 18.9.2 into the marched coefficients and proves that $\alpha^2 - \beta^2 = 4ms$. §4–§5 treat the exponent lane and the binade-split emit. §6 records the hemisphere fold. §7 converts the live-degree count into a hardware lower bound. §8 identifies `ducc0`’s m_{lim} with the larger root of its quadratic.

2 Closed form

Write $\alpha = m + s$, $\beta = |m - s|$, $M = \max(m, s)$ and $n = \ell - M$.

Theorem 2.1.

$$d_{m,-s}^\ell(\theta) = (-1)^{m+s} 2^{\varepsilon_m \text{LG2N}(\ell,m)} \left(\sin \frac{\theta}{2}\right)^\alpha \left(\cos \frac{\theta}{2}\right)^\beta P_n^{(\alpha,\beta)}(x), \quad (2.1)$$

$$\text{LG2N}(\ell, m) = \log_2 \sqrt{\frac{(\ell+m)!(\ell-m)!}{(\ell-s)!(\ell+s)!}}, \quad \varepsilon_m = \begin{cases} +1, & m \geq s, \\ -1, & m < s. \end{cases} \quad (2.2)$$

Proof. Edmonds (1957, eq. 4.1.23) writes, for $d_{m'm}^j$ with $a = |m - m'|$, $b = |m + m'|$ and $n = j - (a + b)/2$,

$$d_{m'm}^j(\beta) = \xi \sqrt{\frac{n!(n+a+b)!}{(n+a)!(n+b)!}} \left(\sin \frac{\beta}{2}\right)^a \left(\cos \frac{\beta}{2}\right)^b P_n^{(a,b)}(\cos \beta), \quad (2.3)$$

where $\xi = 1$ if $m \geq m'$ and $\xi = (-1)^{m-m'}$ otherwise.

Substitute $m' \mapsto m$ and $m \mapsto -s$. Then $a = \alpha$, $b = \beta$, and $a + b = 2M$, so $n = \ell - M$. Unless $m = s = 0$ one has $-s < m$, hence $\xi = (-1)^{-s-m} = (-1)^{m+s}$. The factorial ratio splits on the two branches of M :

- if $m \geq s$, then $n = \ell - m$, $n + a + b = \ell + m$, $n + a = \ell + s$, $n + b = \ell - s$, so the ratio is $2^{+\text{LG2N}}$;
- if $m < s$, then $n = \ell - s$, $n + a + b = \ell + s$, $n + a = \ell + m$, $n + b = \ell - m$, so the ratio is $2^{-\text{LG2N}}$.

This is the ε_m flip in (2.2). □

The kernel writes $(-1)^m$ for $(-1)^{m+s}$. The two agree for even spin (0 and 2, the only spins the march serves). A spin-1 or spin-3 march must restore $(-1)^s$. The closed form *is* Wigner’s function: with $d_{m,-s}^j$ written as Wigner’s explicit sum (Wigner 1931; Sakurai 3.8.33), both sides are sums over $0 \leq k \leq j - M$, the powers of $\sin \frac{\theta}{2}$ and $\cos \frac{\theta}{2}$ match term by term ($\frac{x-1}{2} = -\sin^2 \frac{\theta}{2}$, $\frac{x+1}{2} = \cos^2 \frac{\theta}{2}$), and the coefficients agree by $\binom{j+s}{j-m-k} \binom{j-s}{k} (j-s-k)! (j-m-k)! (k+m+s)! k! = (j+s)! (j-s)!$ for $m \geq s$ (and $(j+m)! (j-m)!$ for $m < s$), which is where the ε_m flip comes from. Numerically, spins 0–3, $\ell \leq 12$, four colatitudes: worst relative difference 6×10^{-13} .

Lean: `Indices.alpha_add_beta`, `alpha_sub_beta`, `alpha_sq_sub_beta_sq`, `jratio_of_ge`, `jratio_of_lt`, `sign_of_even_spin`, `sign_of_odd_spin`; `WignerJacobi.wignerD` (Wigner’s sum), `jacobiForm`, `coeff_ge`, `coeff_lt`, `wignerD_eq_jacobiForm`.

3 Recurrence

DLMF 18.9.2 for $P_n^{(\alpha, \beta)}$, with $S = 2n + \alpha + \beta$, is

$$2n(n + \alpha + \beta)(S - 2)P_n = (S - 1)[S(S - 2)x + (\alpha^2 - \beta^2)]P_{n-1} - 2(n + \alpha - 1)(n + \beta - 1)SP_{n-2}. \quad (3.1)$$

Write $\text{den} = 2n(n + \alpha + \beta)(S - 2)$. Then $v_\ell = (c_1x + c_0)v_{\ell-1} - c_bv_{\ell-2}$ with

$$c_1 = \frac{(S - 1)S(S - 2)}{\text{den}}, \quad c_0 = \frac{(S - 1)(\alpha^2 - \beta^2)}{\text{den}}, \quad c_b = \frac{2(n + \alpha - 1)(n + \beta - 1)S}{\text{den}}. \quad (3.2)$$

Lemma 3.1. $\alpha^2 - \beta^2 = 4ms$ for every $m, s \geq 0$.

Proof. $\alpha - \beta = 2 \min(m, s)$ and $\alpha + \beta = 2 \max(m, s)$. $\square$

Thus no coefficient in (3.2) is a difference of large numbers. A float32 (high, low) split therefore carries them to 2^{-48} .

Seeds: $P_0 = 1$, and DLMF 18.5.7 at $n = 1$ gives

$$P_1^{(\alpha, \beta)}(x) = \frac{\alpha + \beta + 2}{2}x + \frac{\alpha - \beta}{2}. \quad (3.3)$$

For $s = 2$ and $m \geq 2$ this is $(m + 1)x + 2$.

Lean: `JacobiSum.jacobi` (the DLMF 18.5.7 sum), `tj` (its k -th summand in $u = \frac{x-1}{2}$), the ratio identities `tj_succ`, `tj_pred_v`, `tj_pred_u` and their combinations `tj_pred`, `tj_pred2`, the telescoping certificate `W` with `summand_telescopes`, `W_zero`, `W_top`, and DLMF 18.9.2 itself, `jacobi_three_term`; `Recurrence.jacobi_zero`, `jacobi_one`, `c0_eq`, `den_pos`, `recurrence_form`, `march`, `march_eq_jacobi`. No sorry in the chain.

Proof of DLMF 18.9.2 for the sum (as formalised). Let $t_k = \binom{n+\alpha}{n-k} \binom{n+\beta}{k} u^k (u+1)^{n-k}$ for $0 \leq k \leq n$ and $t_k = 0$ otherwise. From $\binom{N}{j+1}(j+1) = \binom{N}{j}(N-j)$ and $\binom{N}{j}(N+1) = \binom{N+1}{j}(N+1-j)$:

$$\begin{aligned} (\alpha + k + 1)(k + 1)(u + 1)t_{k+1} &= (n - k)(n + \beta - k)ut_k, \\ (n + 1 + \alpha)(n + 1 + \beta)(u + 1)t_k^n &= (n + 1 - k)(n + 1 + \beta - k)t_k^{n+1}, \\ (n + 1 + \alpha)(n + 1 + \beta)ut_k^n &= (\alpha + k + 1)(k + 1)t_{k+1}^{n+1}, \end{aligned}$$

and the difference of the last two expresses t_k^n through t_k^{n+1} , t_{k+1}^{n+1} with no division. Applying it twice writes the recurrence's summand at degree N , times $D' = (N + \alpha)(N + \beta)(N + \alpha - 1)(N + \beta - 1)$, through t_k, t_{k+1}, t_{k+2} alone, and it equals $W(k + 1) - W(k)$ for $W(k) = -q_0(k)t_k + (q_2(k - 1) - C(u + 1)(k + 1)(k + 1 + \alpha))t_{k+1}$ (explicit polynomials q_0, q_2, C , found with `sympy` and checked by `ring`); $W(0) = 0$ by the first identity at $k = 0$ and $W(N + 1) = 0$ because t vanishes past N , so the sum telescopes to zero.

4 Exponent carry

Every lane stores $v \cdot 2^{ex}$ with v float32 and $ex \in \mathbb{Z}$. Initially $v = 2^{f - \lfloor f \rfloor} \in [1, 2)$ and $ex = \lfloor f \rfloor$, where

$$f = \alpha \log_2 \sin \frac{\theta}{2} + \beta \log_2 \cos \frac{\theta}{2}. \quad (4.1)$$

Whenever $\max(|v_\ell|, |v_{\ell-1}|)$ leaves $[2^{-24}, 2^{24}]$, both lanes are multiplied by $2^{\mp 24}$ and ex moves by ± 24 .

Lemma 4.1. *The rescaling is exact and commutes with the recurrence.*

Proof. Multiplying a binary float by a power of two changes only the exponent field, hence is exact in the absence of overflow or underflow. With $|v| \leq 2^{24}$ times the growth of one step, and the float32 range $2^{\pm 126}$, overflow before the guard would require growth $> 2^{100}$. From (3.2) one has $|c_1|, |c_0|, |c_b| = O(m)$, so growth per step is $\leq 2^{15}$ at $m < 2^{14}$. The recurrence is linear and homogeneous in $(v_\ell, v_{\ell-1})$, so a common scale persists. $\square$

The band 2^{-24} is chosen so that $v \cdot r$ with $|r|$ of class 2^{24} and a 256-lane sum stays inside float32.

Lean: `ExponentCarry.lane_rescale`, `guard_large`, `guard_small`, `step_homogeneous`, `rec2_scaled` / `march_eq_rec2` / `march_scaled` (the march from seeds scaled by λ is λ times the march at every degree), `no_overflow_before_guard`.

5 Emit

At degree ℓ the program holds (v_θ, ex_θ) on 256 lanes and reduces $\sum_\theta v_\theta 2^{ex_\theta - e_{\max}} r_c(\theta)$ with $e_{\max} = \max_\theta ex_\theta$. Lanes with $ex_\theta - e_{\max} < -126$ are flushed to zero.

Lemma 5.1 (flush bound). *Flushed lanes change the tile partial by less than 2^{-70} relative to the largest lane's term.*

Proof. Stored $|v|$ lie in $[2^{-24}, 2^{24}]$. The largest term is at least $2^{-24} \cdot 2^{e_{\max}} \cdot |r|$; a flushed term is at most $2^{24} \cdot 2^{e_{\max} - 127} \cdot |r|$. The ratio is 2^{-79} , and there are at most 2^8 lanes. $\square$

From (2.1) the tile partial of (1.1) is

$$A_c^{\text{tile}}(\ell, m) = (-1)^{m+s} 2^{\varepsilon \text{LG2N} + e_{\max}} \sum_\theta [v_\theta 2^{ex_\theta - e_{\max}}] r_c(\theta). \quad (5.1)$$

Write $\varepsilon \text{LG2N} = \text{lex} + \text{frac}$ with $\text{lex} = \lfloor \varepsilon \text{LG2N} \rfloor$ and $\text{frac} \in [0, 1)$. The kernel stores

$$p = \text{fl}_{32} \left(\text{fl}_{32} \left(\sum_\theta \text{val}_\theta r_c(\theta) \right) \cdot \text{fl}_{32} \left((-1)^{m+s} 2^{\text{frac}} \right) \right), \quad e = \text{lex} + e_{\max}, \quad (5.2)$$

and the driver returns $\text{fl}_{64}(p) \cdot 2^e$, summing tiles in float64.

Proposition 5.2. $\text{fl}_{64}(p) 2^e$ equals (5.1) with relative error at most $(1+u)^2(1+\delta) - 1$, where $u = 2^{-24}$ and δ is the relative error of the float32 tree sum. Multiplication by 2^e is exact for $-1022 \leq e \leq 1023$. A partial with $e < -1022$ is below 2^{-1022} in magnitude and is dropped.

Proof. 2^e is exactly representable in float64 on that range, and multiplication by it only changes the exponent field. The two float32 roundings are $\text{fl}_{32}(2^{\text{frac}})$ and the product; each contributes a factor in $[1-u, 1+u]$. The tree error δ is the same term the previous float64-emit kernel carried. Harmonic coefficients are $O(1)$ sums of these partials, so a contribution below 2^{-1022} is below float64 resolution. $\square$

Measured relative change against the float64-emit kernel, over every lane: 7.5×10^{-8} , 7.3×10^{-8} , 6.6×10^{-8} at $N_{\text{side}} = 1024, 2048, 4096$, i.e. $\sim 2u$ as predicted.

Lean: `Emit.flushed_lane_small`, `flush_bound`, `binade_split`, `frac_factor_bounds`, `int_binade_is_zp`, `two_roundings`, `dropped_partial_small`.

6 Hemisphere fold

Lemma 6.1. $d_{m,0}^\ell(\pi - \theta) = (-1)^{\ell+m} d_{m,0}^\ell(\theta)$.

Proof. $d_{m,0}^\ell(\theta) = \sqrt{(\ell-m)!/(\ell+m)!} P_\ell^m(\cos \theta)$ and $P_\ell^m(-x) = (-1)^{\ell+m} P_\ell^m(x)$ (DLMF 14.7.17). Equivalently, $P_n^{(\alpha,\alpha)}(-x) = (-1)^n P_n^{(\alpha,\alpha)}(x)$ from the explicit sum, and $\alpha = \beta = m$ at spin 0. $\square$

HEALPix rings satisfy $\theta_{N-1-i} = \pi - \theta_i$ and share weight and ϕ_0 . The sum (1.1) therefore collapses, per northern lane, to

$$\sum_{\text{north}} d_i [G_i + (-1)^{\ell+m} G'_i]. \quad (6.1)$$

One recurrence, two right-hand sides; the parity sign is applied outside the kernel. This halves the lane count for spin 0.

For spin 2 the code uses $T[m, \pi - \theta, \ell] = (-1)^{\ell-m'} T[-m, \theta, \ell]$ with $m' = -s$ (measured residual 2.3×10^{-12}). It is a consequence of the general reflection $P_n^{(\alpha, \beta)}(-x) = (-1)^n P_n^{(\beta, \alpha)}(x)$, which swaps the two half-angle powers in (2.1): for $s \leq m \leq \ell$, $d_{m, -s}^\ell(\pi - \theta) = (-1)^{\ell+m} d_{m, s}^\ell(\theta)$, and $d_{-m, -s}^\ell = (-1)^{m+s} d_{m, s}^\ell$ is read off the same Jacobi form. Lean: `Mirror.jacobi_reflect`, `rowNeg_reflect`, `mirror_identity`.

Lean: `Fold.jacobi_symmetric_parity`, `row0_reflect`, `sum_pairs_even`, `sum_pairs_odd`, `fold_even`.

7 The arithmetic wall

Lemma 7.1. $\sum_{m < L} (L - m) = L(L + 1)/2$.

Let N be the number of (m, ℓ, θ) triples visited. Before polar skip, $N = \frac{1}{2}L(L + 1) \lceil n_\theta/2 \rceil$ on the folded spin-0 route and $N = \frac{1}{2}L(L + 1) n_\theta$ at spin 2.

N_{side}	L	rings (spin 0 / 2)	N spin 0	N spin 2
1024	3072	2048 / 4095	9.7×10^9	1.9×10^{10}
2048	6144	4096 / 8191	7.7×10^{10}	1.5×10^{11}
4096	12288	8192 / 16383	6.2×10^{11}	1.2×10^{12}

An analysis pass with n_c channels costs at least $(2 + n_c)N$ fused multiply-adds. On this card the measured float64 FMA rate is $R_{64} = 0.45 \times 10^{12} \text{ s}^{-1}$ (datasheet 0.94×10^{12}). Hence any float64 SIMT spin-0 analysis at $N_{\text{side}} = 4096$ needs

$$T \geq \frac{6 \cdot 6.2 \times 10^{11}}{0.94 \times 10^{12}} \text{ s} = 4.0 \text{ s} \quad (7.1)$$

at the datasheet rate (8.3 s measured), against ducc0's 2.02 s on the host. No float64 kernel on this card can beat the CPU reference at the top of the board.

Lean: `Wall.live_degrees`, `time_lower_bound`.

8 Polar skip

For $m > (\ell + \frac{1}{2}) \sin \theta$ the function $d_{m, -s}^\ell(\theta)$ is in its forbidden region. Writing $u = \sqrt{\sin \theta} d_{m, 0}^\ell$,

$$u'' + \left[\left(\ell + \frac{1}{2} \right)^2 - \frac{m^2 - \frac{1}{4}}{\sin^2 \theta} \right] u = 0. \quad (8.1)$$

With $\ell' = \ell + \frac{1}{2}$ the WKB action from the turning point θ_t ($\sin \theta_t = m/\ell'$) down to $\theta < \theta_t$ is

$$S = \int_\theta^{\theta_t} \sqrt{\frac{m^2}{\sin^2 t} - \ell'^2} dt \sim \frac{2\sqrt{2}}{3} \frac{\delta^{3/2}}{\sqrt{\ell' \sin \theta} \cos \theta}, \quad (8.2)$$

where $\delta = m - \ell' \sin \theta$. This estimate is not formalised.

ducc0 neglects, on the ring at θ , every order above

$$m_{\text{lim}} = s |\cos \theta| + \sqrt{(\ell_{\text{max}} \sin \theta + \text{ofs})^2 - s^2 \sin^2 \theta}, \quad \text{ofs} = \max(100, 0.01 \ell_{\text{max}}). \quad (8.3)$$

Lemma 8.1. (8.3) is the larger root of $m^2 - 2s |\cos \theta| m + s^2 - t_1^2 = 0$ on a ring with $\cos^2 \theta + \sin^2 \theta = 1$ and $t_1 = \ell_{\text{max}} \sin \theta + \text{ofs}$. For $s = 0$ one has $m_{\text{lim}} = \ell_{\text{max}} \sin \theta + \text{ofs}$.

Every $\ell \leq \ell_{\max}$ then has $\delta \geq \text{ofs}$. At $\ell_{\max} = 12287$, $\sin \theta = \frac{1}{2}$, one finds $S \sim 19$, so neglected values sit below $\sim 10^{-8}$ times the oscillatory amplitude. The march skips a tile when $m > \max_{\text{tile}} m_{\text{lim}}(\theta)$. Because the neglected set is the reference's own, agreement with `ducc0` is unchanged to the printed digit.

N_{side}	analysis (tile 256)	synth. spin 0 (1024)	synth. spin 2 (512)
1024	13.9%	2.5%	9.3%
2048	17.5%	10.0%	14.8%
4096	19.3%	15.1%	17.9%

Lean: `PolarSkip.mlim_is_root, mlim_ge, mlim_spin_zero, skipped_degrees`.

A Lean correspondence

Build: `cd formal/lean && ~/.elan/bin/lake build`. There is no `sorry`; `lake env lean` `CheckAxioms.lean` prints the axioms of every main theorem (`propext`, `Classical.choice`, `Quot.sound`).

Section	File	Content
§2	<code>Indices.lean</code>	$\alpha + \beta = 2M$, $\alpha^2 - \beta^2 = 4ms$, both factorial branches, sign
§2	<code>WignerJacobi.lean</code>	Wigner's explicit sum equals the Jacobi closed form, both branches
§3	<code>JacobiSum.lean</code>	Jacobi sum; DLMF 18.9.2 by summand ratios and a telescoping certificate
§3	<code>Recurrence.lean</code>	seeds, rearrangement, <code>march = jacobi</code>
§4	<code>ExponentCarry.lean</code>	exact rescaling, homogeneous step, no overflow before the guard
§5	<code>Emit.lean</code>	flush 2^{-71} , binade split, $[1, 2)$ mantissa, two roundings
§6	<code>Fold.lean</code>	$P_n^{(\alpha, \alpha)}(-x) = (-1)^n P_n^{(\alpha, \alpha)}(x)$, paired rings
§6	<code>Mirror.lean</code>	$P_n^{(\alpha, \beta)}(-x) = (-1)^n P_n^{(\beta, \alpha)}(x)$; the spin-2 mirror identity
§7	<code>Wall.lean</code>	Lemma 7.1; time bound from an FMA count and a rate
§8	<code>PolarSkip.lean</code>	Lemma 8.1; degrees removed by a cutoff

Not formalised: the WKB action (8.2).

References

- [1] A. R. Edmonds, *Angular Momentum in Quantum Mechanics*, Princeton University Press, 1957.
- [2] NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/>.